Kapasa Makasa University

Department	**: ICT**
Course	**: Cybersecurity Year: 3**
Course Name	**: System Vulnerability Assessment**
Course Code	**: 330**
Academic year	**: 2024**
Task	**: Test 1**
Date	**: 10th May 2024**
Duration	**: Two (2) Hours**
Lecturer	**: Paul M. Mutunda**

Instructions

1. Write your **Name** and **Student Number** on all the answer sheets you will use in this test
2. Total Marks [ ] for this paper is **100**.
3. Time allowed is **Two (2) Hours**
4. There are **two Sections** in this paper, Section **A** and **B**. Section A comprises of **One (1) question** carrying **25 Marks** and Section B comprises of **Three (3) questions**, each carrying **25 Marks**. Attempt **ALL** questions.
5. Neat and clear written work is encouraged

SECTION A [One word answer – 25 Marks]

1. Another name for cybersquatting … [1]
2. Acronym SDE stands for … [1]
3. This is when we replace sensitive data with fictional values for nonproduction environments [1]
4. This is a damage caused when a threat exploits a vulnerability … [1]
5. Compromised data confidentiality [1]
6. Something that can exploit a vulnerability … [1]
7. An update released to address specific issues in a software [1]
8. A flaw or weakness in a system or application that occurs due to improper setup among others … [1]
9. An action taken to correct a vulnerability or damage to a system [1]
10. Another name for zero day … [1]
11. This is when a hacker or an attacker attempts to guess a user's login credentials manually without using any software … [1]
12. …… involves a systematic attempt to access different directories by guessing their names or simply enumerating through a list of common file names [1]
13. Occur when flaws or weaknesses are present in the manufacturer's codes which allow attackers to exploit them for malicious purposes [1]
14. A framework of policies and technologies to ensure that the right users have access to a technological resource. [1]
15. The use of attackers against an enemy causing comparable harm or disruption to important computer systems. … [1]
16. Another name for vulnerability scanning … [1]
17. A condition of a vulnerability being severe is called … [1]
18. A term that describes incomplete installations, poorly executed system changes, and default deployments on a computer system. [1]
19. …. Occur when a vulnerability is flagged that doesn't even exists [1]
20. SQLi is an example of … [1]
21. A methodology used to measure the scope of the assessment is known as …… while ….. is pertains to the granularity of the examination conducted on each asset. [2]
22. These are software and operating system updates that address security vulnerabilities within a software or product [1]
23. A term used to describe a process of automatically extracting information from websites using web crawlers. [1]
24. A process of defining, identifying, classifying and prioritizing security weaknesses such as servers, operating systems, application software and network infrastructure is called … [1]

Section B [75 Marks]

25. Vulnerability assessment process is an ongoing process.
    (a) Identify the four (4) steps involved in carrying out this process [4]
    (b) With the aid of a diagram, explain the identified steps in (a) [11]
    (c) Give five (5) differences between vulnerability management and vulnerability assessment [10]

26. Explain the following in detail, giving the impact and mitigation of the impacts.
    (a) Hardware vulnerabilities [5]
    (b) Organisational vulnerabilities [5]
    (c) Personal vulnerabilities [5]
    (d) Software vulnerabilities [5]
    (e) Network vulnerabilities [5]

27. As an IT Chief of Security, you conducted a scan on a Local Area Network (LAN) for your organisation called "Nested" for a period of 3 days. The organisation comprises of 102 employees and has about 66 computers connected. Your start date was 10th May 2024 and expected date of completion was 13th May 2024. The industry of the organisation concerns Accounting. The following are some of the findings in your system auditing:
    - Malware and botnet events detected – 24
    - Evasive applications – 12
    - Filesharing applications – 14
    - Bandwidth consuming applications – 11
    - Security risk applications – High
    - Application vulnerability exploit – High
    - Malware detected – 06

    Task
    Write a simple technical report considering the format of a technical report or 6 vital aspects.